



**Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 • STANDARD 6.AT.6A

# Write Algebraic Expressions

Lesson 6-3 • Teacher Copy

**Name:** \_\_\_\_\_

**Date:** \_\_\_\_\_

**Class:** \_\_\_\_\_



## TODAY'S OBJECTIVES

**Content Objective:** I can write algebraic expressions from words and real-world situations.

**Language Objective:** I can explain my expression using the words **variable**, **coefficient**, **constant**, and **algebraic expression**.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Variable

Algebraic Expression

Coefficient

Constant

Like terms



Tap any word to see what it means and a picture.

1

A letter that stands for an unknown number is a .

2

A math phrase with variables, numbers, and operations but no equal sign is an

.

3

The number multiplied by a variable, like the 7 in  $7n$ , is the

.

4

A term that is just a number and never changes, like the 5 in  $3x + 5$ , is a

.

5

Terms with the same variable part are .

## Write About the Math

### Guided ESOL writing

Use the support level you need. Every level answers the same math question.

### 1. Understand the Question

#### COMPARE

**How do you turn words into an expression? Why does '12 less than a number  $g$ ' become  $g - 12$  and not  $12 - g$ ?**

**Your job:** Answer the question. Use math evidence. Explain how the evidence proves your answer.

*Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.*

### 2. Plan Your Math Words

Check at least two words you will use.

☐ **Variable** — A letter that stands for a number that is unknown or can change.  
*Variable — Una letra que representa un número desconocido o que puede cambiar.*

☐ **Algebraic Expression** — A math phrase that has at least one letter.  
*Expresión algebraica — Una frase matemática que tiene al menos una letra.*

☐ **Coefficient** — The number in front of a letter, like the 3 in  $3x$ .  
*Coficiente — El número frente a una letra, como el 3 en  $3x$ .*

☐ **Constant** — A number on its own that does not change.  
*Constante — Un número solo que no cambia.*

☐ **Like terms** — Terms with the same letter, like  $2x$  and  $5x$ .  
*Términos semejantes — Términos con la misma letra, como  $2x$  y  $5x$ .*

**Say it first:** Say your idea to a partner or quietly to yourself before you write.

*Di tu idea a un compañero o en voz baja antes de escribir.*

**The words '\_\_\_\_' mean \_\_\_\_\_, so the expression is \_\_\_\_\_.**

*Las palabras '\_\_\_\_' significan \_\_\_\_\_, así la expresión es \_\_\_\_\_.*

### 3. Build Your Explanation

**Model the parts:** This model shows the parts of an explanation. It does not answer your writing question.

**Claim:** Write  $12t + 50$ : the  $12t$  grows with tickets, the  $50$  stays fixed.

**Evidence:** Students write  $12t + 50$ , identify  $t$  as the variable number of tickets,  $12$  as the coefficient (price per ticket), and  $50$  as the constant venue fee that does not change.

**Reasoning:** This evidence connects the definition of algebraic expression to the mathematical claim.

#### Start

Most language support

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The words '\_\_\_' mean \_\_\_, so the expression is \_\_\_.

*Las palabras '\_\_\_' significan \_\_\_, así la expresión es \_\_\_.*

- It is  $g - 12$ , not  $12 - g$ , because 'less than' means \_\_\_.

*Es  $g - 12$ , no  $12 - g$ , porque 'menos que' significa \_\_\_.*

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#### Build

Some language support

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_.

*Mi afirmación es \_\_\_.*

- I know because \_\_\_.

*Lo sé porque \_\_\_.*

- So \_\_\_.

*Entonces \_\_\_.*

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## Explain

### Light language support

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

- **My claim is \_\_\_\_.**  
*Mi afirmación es \_\_\_\_.*
- **My math evidence is \_\_\_\_.**  
*Mi evidencia matemática es \_\_\_\_.*
- **This proves \_\_\_\_ because \_\_\_\_.**  
*Esto demuestra \_\_\_\_ porque \_\_\_\_.*

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## 4. Check Your Explanation

- ☐ I answered the question.  
*Respondí la pregunta.*

- ☐ I used math evidence: numbers, an equation, a model, or a comparison.  
*Usé evidencia matemática: números, una ecuación, un modelo o una comparación.*

- ☐ I explained how the evidence proves my answer.  
*Explicué cómo la evidencia demuestra mi respuesta.*

- ☐ I used at least two math words.  
*Usé por lo menos dos palabras matemáticas.*

- ☐ I reread complete sentences and punctuation.  
*Volví a leer mis oraciones completas y la puntuación.*

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## Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

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## Answer Key & Teacher Guide

1. **Notes 1:** Variable
2. **Notes 2:** Algebraic Expression
3. **Notes 3:** Coefficient
4. **Notes 4:** Constant
5. **Notes 5:** Like terms
6. **Watch for this mistake:** *A common mistake in Write Algebraic Expressions is treating 'each' or 'per' as if it means add instead of multiply, and attaching the variable to the wrong number. For 'a \$25 setup fee plus \$8 per hour, h,' students often write  $25 + 8 + h$  (adding all three numbers) or  $25h + 8$  (multiplying the flat fee by  $h$ ) instead of the correct  $8h + 25$ . Since \$8 applies per hour, it must multiply the variable ( $8h$ ), and the flat \$25 fee is a constant added only once — check that the per-unit rate, not the flat fee, is the number multiplied by the variable.*
7. **Try It 1:** A.  $x - 5$  — 'Less than' subtracts from the number: start with  $x$  and take away 5, so  $x - 5$ .
8. **Try It 2:** A.  $2s$  — 'Twice' means two times the number, so twice  $s$  is  $2s$ .